

Experiment ‘p_6_2_d_replace_with_larger_neighbor_seq’

Results

June 4, 2026

Experiment outcome: SUCCESS
Bad responses: 0
Responses containing assume : 0
Responses containing expect : 0
Resolution attempts: 2
Hard fails (resolution): 0
Soft fails (resolution): 0
Verification attempts: 2

Problem Specification

Problem name: p_6_2_d_replace_with_larger_neighbor_seq

Natural language statement: Write a method to replace each element of an array, except the first and last, by the larger of its two neighbors.

Method signature: p_6_2_d_replace_with_larger_neighbor_seq(inputs: seq<int>) returns (result: seq<int>)

Ensures

- `|result| == |inputs|`
- `result[0] == inputs[0]`
- `result[|inputs| - 1] == inputs[|inputs| - 1]`
- forall `i :: 1 <= i < |inputs| - 1 ==> result[i] == if inputs[i - 1] >= inputs[i + 1] then inputs[i - 1] else inputs[i + 1]`

Requires

- `|inputs| >= 3`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to replace each element of an array, except the first and last, by the
→ larger of its two neighbors.

The signature should be:

p_6_2_d_replace_with_larger_neighbor_seq(inputs: seq<int>) returns (result: seq<int>)

The method should respect the following contract:

requires |inputs| >= 3, ensures |result| == |inputs|, ensures result[0] == inputs[0],
→ ensures result[|inputs| - 1] == inputs[|inputs| - 1], ensures forall i :: 1 <= i <
→ |inputs| - 1 ==> result[i] == if inputs[i - 1] >= inputs[i + 1] then inputs[i -
→ 1] else inputs[i + 1]

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn0FF5zVnl83slXPx3ijTkTUxfbvR

Model: gpt-5.5-2026-04-23

Created at: 1780571385

Finish reason: stop

Usage: 466 tokens in, and 612 tokens out

```
{  
  result := seq(|inputs|, i =>  
    if i == 0 then inputs[0]  
    else if i == |inputs| - 1 then inputs[|inputs| - 1]  
    else if inputs[i - 1] >= inputs[i + 1] then inputs[i - 1]  
    else inputs[i + 1]);  
}
```

Program → GenAI

When using dafny verify, the below error is emitted and verify fails:

problem_data/p_6_2_d_replace_with_larger_neighbor_seq/program.dfy(11,18): Error: index
→ out of range
|
11 | else if inputs[i - 1] >= inputs[i + 1] then inputs[i - 1]

```

| ~
problem_data/p_6_2_d_replace_with_larger_neighbor_seq/program.dfy(11,35): Error: index
  ↳ out of range
|
11 | else if inputs[i - 1] >= inputs[i + 1] then inputs[i - 1]
| ~

```

Dafny program verifier finished with 1 verified, 2 errors

Correct the error in code by altering only the method body. Produce and show only the
 ↳ Dafny body, including the curly braces that surround it. Do not show the signature
 ↳ nor contract. You must not use 'assume' nor 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null
ID: chatcmpl-Dn0Fr9hgJEiZ1tgde7ROn5tXFEBFG
Model: gpt-5.5-2026-04-23
Created at: 1780571423
Finish reason: stop
Usage: 764 tokens in, and 985 tokens out

```

{
  result := seq(|inputs|, i =>
    if i == 0 then inputs[0]
    else if i == |inputs| - 1 then inputs[|inputs| - 1]
    else if 1 <= i < |inputs| - 1 then
      if inputs[i - 1] >= inputs[i + 1] then inputs[i - 1] else inputs[i + 1]
    else inputs[0]);
}

```

Final Program

```

method p_6_2_d_replace_with_larger_neighbor_seq(inputs: seq<int>) returns (result: seq<
  ↳ int>)
  requires |inputs| >= 3
  ensures |result| == |inputs|
  ensures result[0] == inputs[0]
  ensures result[|inputs| - 1] == inputs[|inputs| - 1]
  ensures forall i :: 1 <= i < |inputs| - 1 ==> result[i] == if inputs[i - 1] >=
    ↳ inputs[i + 1] then inputs[i - 1] else inputs[i + 1]
{
  result := seq(|inputs|, i =>
    if i == 0 then inputs[0]
    else if i == |inputs| - 1 then inputs[|inputs| - 1]
    else if 1 <= i < |inputs| - 1 then
      if inputs[i - 1] >= inputs[i + 1] then inputs[i - 1] else inputs[i + 1]
    else inputs[0]);
}

```

}

Total Token Usage

Input tokens: 1230

Output tokens: 1597

Reasoning tokens: 1371

Sum of ‘total tokens’: 2827

Experiment Timings

Overall Experiment started at 1780571384834, ended at 1780571473607, lasting 88773ms (88.77 seconds)

Iteration #1 started at 1780571384841, ended at 1780571422722, lasting 37881ms (37.88 seconds)

Iteration #2 started at 1780571422722, ended at 1780571473607, lasting 50885ms (50.89 seconds)